Arpon Kapuria

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Education

National Institute of Technology Tiruchirappalli — Tiruchirappalli, India
B.Tech in Computer Science & Engineering — **November 2020 – June 2024**

Courseworks: Data Structures and Algorithms · Operating Systems · Database Management Systems · Computer Networks · Software Engineering · Machine Learning · Artificial Intelligence · NLP · Image Processing · Augmented and Virtual Reality · Technical Writing

Experience

Advanced Machine Intelligence Research Lab — Dhaka, Bangladesh
Research Assistant | *Advisor: Prof. M. Firoz Mridha* — **November 2025 – Present**

- Engaged in research on Generative NLP and assisted in implementing deep learning models.

Indian Institute of Technology Bombay — Mumbai, India
Research Intern | MeDAL Lab | *Advisor: Prof. Amit Sethi* — **May 2023 – July 2023**

Project 1: Enhancing Self Supervised Learning framework - BYOL

- Conducted an in-depth survey on self-supervised learning frameworks and reproduced BYOL and simCLR in **PyTorch** to establish a baseline for improvements.
- Introduced changes to BYOL by incorporating a novel loss function, algorithmic modifications and architectural adjustments to improve representation learning.

Project 2: Radiology DICOM Image Anonymization

- Created a user-friendly **Flask** based system for DICOM image processing and anonymization using Pydicom, ensuring HIPAA-compliant privacy across **5,000+ medical scans**.
- Evaluated the CRAFT model for text detection in X-ray images and extracting clinically relevant annotations.

National Institute of Technology Tiruchirappalli — Tiruchirappalli, India
Undergraduate R&D Assistant | Industrial Automation Lab — **October 2022 – February 2023**

Project: Assam Smart Home Automation — *Advisors: Prof. M. Brindha, Prof. G. S. Ilango*

- Developed a **Flutter** app to automate energy operations and digital billing for a solar-powered microgrid in Assam, India, serving **200+ households**.
- Integrated Flutter frontend with Django-based REST API and deployed an MQTT broker enabling real-time data exchange with **50+ IoT sensors** for energy consumption and control.
- Project supported by the Ministry of Science & Technology, Government of India, under the initiative **SUSTENANCE** to achieve Novel Carbon Neutral Energy Communities.

Skills

- **Programming Languages:** Python, C, C++, SQL, Bash
- **AI / ML:** PyTorch, Hugging Face, LangChain, LangGraph, vLLM, MLflow, DVC
- **Backend & Data:** FastAPI, Ray (Data, Serve), PostgreSQL, Redis, Vector DB
- **Infrastructure & MLOps:** Git, Docker, Kubernetes, Terraform, AWS (S3, EKS, IAM, VPC), CI/CD
- **Observability:** Prometheus, Grafana, OpenTelemetry
- **Spoken Languages:** Bengali (Native), English (IELTS 7.5), Hindi (Conversational), German (A1.1)

Open Source Contributions

- Huggingface/transformers
- LangChain/langchain
- PyTorch/pytorch

Selected Projects

Agentic RAG System with Hybrid Retrieval & Scalable Deployment **March 2026 – April 2026**
Python, FastAPI, Ray, LangGraph, vLLM, Qdrant, Neo4j, PostgreSQL, Docker, Kubernetes, Terraform, AWS, Prometheus, Grafana, OpenTelemetry

- Built an end-to-end **Agentic RAG pipeline** with persistent conversational memory, supported by a distributed ingestion pipeline using **Ray Data** and event-driven **S3** ingestion to transform raw documents into embeddings and knowledge-graph structures.
- Combined vector search (**Qdrant**) with graph traversal (**Neo4j**) and orchestrated it via **LangGraph** with HyDE-based query rewriting, planner-driven routing, and tool execution for entity-aware, multi-hop reasoning beyond embedding-only retrieval.
- Provisioned infrastructure using **Terraform** on **AWS** (VPC, EKS, S3, IAM) and deployed LLM inference via **Ray Serve** (**vLLM** backend) with autoscaling, semantic caching, observability (**Prometheus, Grafana, Opentelemetry**), and evaluation (**RAGAS, LLM-as-judge**) to maintain system reliability and answer faithfulness under load.

Cold Email Generator **April 2025**
Python, LangChain, FAISS, RAGAS, Llama-4, BeautifulSoup (BS4)

- Automated an end-to-end multimodal RAG system to streamline graduate application outreach by aligning applicant profiles with professor research via web scraping, text parsing, and CV image analysis, achieving a **92.6% relevance score** (RAGAS).
- Designed a LangChain-based pipeline leveraging **LLaMA-4 Maverick (Groq AI)**, **Jina Embeddings v3**, and **FAISS**, leveraging multimodal embeddings to generate contextually personalized cold emails.
- Incorporated **Cohere Reranker v3.5**, achieving a **Precision@K of 87.3%**, with robustness validated across diverse input formats (PDFs, HTML, images).

Malicious Website Detection Using Machine Learning **October 2022 – November 2022**
Python, Scikit-Learn, Flask, Docker, JavaScript, Chrome Extension tools

- Implemented and deployed a machine-learning model via **Flask** and **Docker**, achieving **94% accuracy** for real-time malicious website detection.
- Trained and benchmarked **4 models** — KNN, SVM, LR & MLP — and optimized hyperparameters, resulting **14% enhancement** in model performance .
- Built a Chrome extension using JavaScript to extract **27 real-time features** (e.g., URL length, domain authority) from webpages and integrated user feedback collection, used by **100+ test users**.
- Applied a continual learning pipeline that retrains the model every **24 hours** on the feedback data, reducing false positives by **almost 6%** and increasing detection robustness.

Achievements

2020	**ICCR Scholarship** – Ministry of External Affairs, Government of India; awarded for academic excellence and funded my bachelor studies.
2017	**Academic Excellence Award** – Chamber of Commerce, Kushtia; SSC Board (10th) Results.
2015	**General Grade Scholarship** – Government of Bangladesh; JSC Board (8th) Results.
2012	**1st Place** – Zilla Shilpakala Academy, Kushtia; Art Competition (Independence Day).
2011	**2nd Place** – Bangladesh Udichi Shilpigoshthi, Kushtia; Art Competition (Bengali New Year 1418).

References

Dr. M. Brindha, Associate Professor
Department of Computer Science & Engineering, NIT Trichy, India | brindham@nitt.edu

Amit Sethi, Professor
Department of Electrical Engineering, IIT Bombay, India | asethi@iitb.ac.in

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